Power Electronics Laboratory
Session 1: DC-DC converter 1

24-03-2025

Objectives

1. Study the steady-state operation of the synchronous buck converter.
2. Study the switching process of a DC-DC power converter.
3. Understand the Pulse Width Modulation.
4. Study the impact of the parameters on the power converter performance.
5. Study the impact of losses on the performance of the power converter.

General instructions

1. Introduction is evaluated with **0.5** points maximum.
2. Conclusion is evaluated with **0.5** points maximum.
3. For every **3** figure format and order error, **0.1** points will be discounted.
4. All figures and equations must be referenced.
5. In many tasks, it is not indicated what variables must be plotted. The student must be able to decide what variables to plot, and how to plot them.

Part 1

Build the power stage shown in Fig. 1 using Plecs simulator. For the semiconductor switches, use single IGBTs with antiparallel diodes.

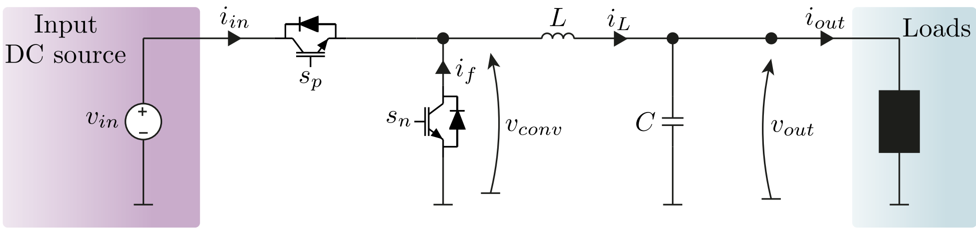

Figure 1: Synchronous Buck converter.

Please, label and measure the following electrical quantities according to the directions shown in Fig. 1.

- Input voltage: v_{in}.
- Input current: i_{in}.
- Current through the neutral-connected IGBT: i_f.
- Converter voltage: v_{conv}.
- Inductor current: i_L.
- Output voltage: v_{out}.
- Output current: i_{out}.

Configure the circuit with the parameters shown in Table 1, and set the initial conditions of the inductor and capacitor to 0 A and 0 V, respectively.

Table 1: System parameters.

Parameter	Value
Input voltage	500V
Inductance	200μH
Capacitance	100μF
Resistive load	5Ω

Part 2

The ideal IGBTs are turned on and off with the gate signals s_p and s_n, which must operate in a complementary manner. For simplicity, they will be considered as digital signals (1 and 0). Therefore $s_n = \overline{s}_p$.

In Plecs, build the diagram shown in Fig. 2, and connect it to the power stage using the appropriate labels. Configure the pulse generator block to obtain a digital signal that oscillates between 0 and 1 at a frequency of 10 kHz.

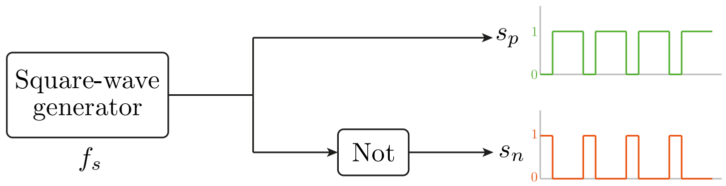

Figure 2: Test gate signal generator.

Tasks:

1. Choose **three** different duty cycle values and check the steady-state operation of the power converter.
2. Calculate the theoretical output voltage for every duty cycle.
3. Does the output voltage match the expected value?
4. What happens if s_p and s_n are activated at the same time?
5. Comment all your results.

Part 3

The pulse generator used in **Part 2** will be replaced with a Pulse-Width Generator (PWM). This component is fundamental in power electronics, and it will be utilized during the remaining laboratory sessions.

The input of this system is the reference duty cycle, d^*, and its output corresponds to the gate pulses s_p and s_n. Two simplified implementations of the PWM are shown in Fig. 3.

Basically, the PWM compares the reference duty cycle with the carrier triangular waveform. **Both quantities must operate in the same range**. When the reference duty cycle is bigger than the carrier, s_p is activated. On the other side, when the reference duty cycle is lower than the carrier, s_p is turned off. Therefore, by increasing the reference duty cycle, the period in which s_p is turned on increases. **In one switching cycle, the average value of s_p is equal to the reference duty cycle**.

Configure the triangular waveform generator as follows:

- Minimum value = 0.
- Maximum value = 1.
- Frequency = 10kHz.
- duty cycle = 0.5 (To obtain a symmetrical triangular waveform).

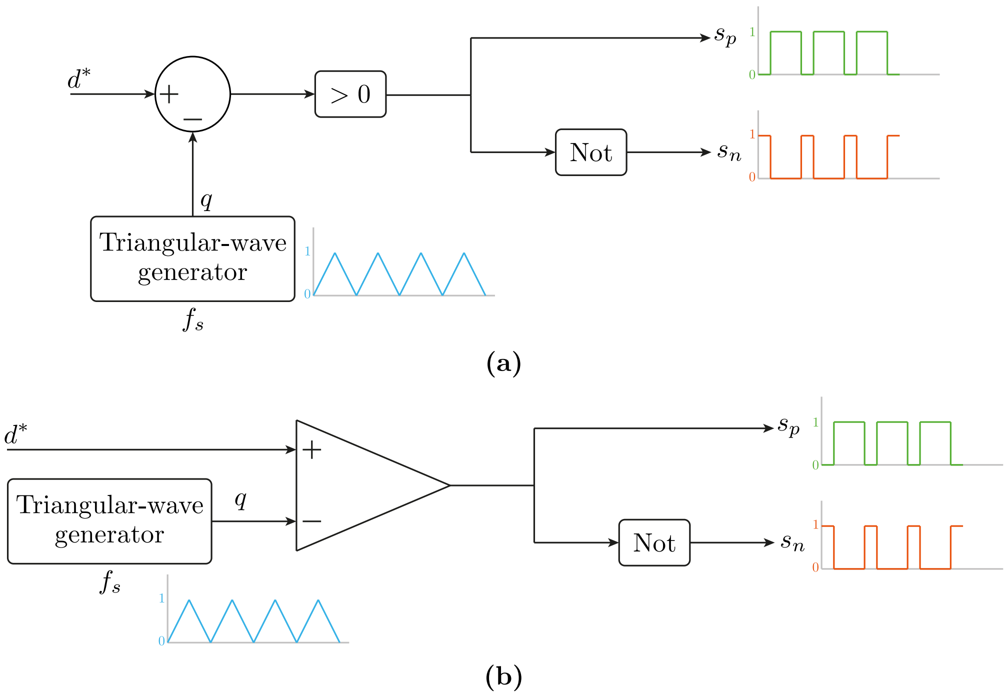

Figure 3: PWM Generators.

Tasks

1. Implement **one** of the PWM generators shown in Fig. 3.
2. Verify the operation of the PWM generator with the duty cycle values used in **Part 2**. Plot the following quantities (3 cycles):
   - Reference duty cycle value and carrier waveform.
   - The positive pulse, s_p.
3. Calculate the average value of s_p. Is it equal to the reference duty cycle? What happens with v_{conv}? **Comment**.
4. Check if the output voltage matches the values obtained in **Part 2**.
5. For each duty cycle value, simulate until reaching the stady-state. Plot 3 cycles of the following variables:
   - Input voltage.
   - The switching signal, s_p.
   - The converter voltage, v_{conv}.
   - The current on both IGBTs.
   - The inductor current.
   - The output voltage.
   - The output current.
6. Comment all your results.

Part 4

The election of the passive components, as well as the switching frequency of the semiconductors are critical for the power converter. Therefore, this part of the laboratory assesses the impact of modifying the parameters of the power converter on the operation of the power converter.

For each one of the following tasks, plot 3 cycles of the following quantities in steady state.

- Input voltage.
- The switching signal, s_p.
- The current on both IGBTs.
- The inductor current.
- The output voltage.
- The output current.

Tasks

1. Simulate the power converter with the parameters of Table 1. Then, increase and decrease the value of the inductance, L.
2. Simulate the power converter with the parameters of Table 1. Then, increase and decrease the value of the capacitance, C.
3. Simulate the power converter with the parameters of Table 1. Then, increase and decrease the value of the switching frequency.
4. Simulate the power converter with the parameters of Table 1. Then, increase and decrease the value of the load resistance.

Please, pay attention to the behaviour of the inductor current and output voltage ripple in each case. **Comment all your results**.

Part 5

The power converter studied in the previous sections is ideal, i.e., it has no losses.

Modify the circuit implemented previously and add the resistances shown in Fig. 4. This is a very simple model that emulates the conduction losses of the semiconductors, the copper losses of the inductor, and the equivalent series resistance (ESR) of the capacitor. Much more complex models can model effects such as source resistances, line models, switching losses of the semiconductors, nonlinearities of the passive components, and parasitic capacitances.

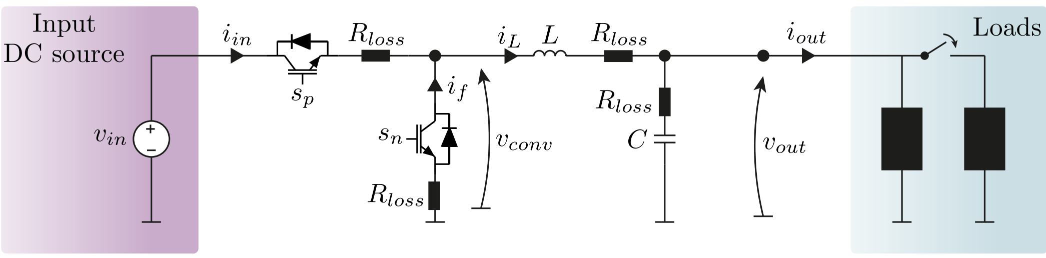

Figure 4: Nonideal Synchronous Buck converter.

Tasks

1. Use the parameters of Table 1 in addition to $R_{loss} = 50m\Omega$.
2. Choose **one** of the values for the duty cycle employed in the previous sections and simulate the system.
3. Compare the output voltage (With and without losses).
4. Adjust manually the duty cycle to obtain the same output voltage as in the ideal case. For the ideal and nonideal cases.
5. With the modified duty cycle obtained in the previous task: Simulate the system, and once the steady state is achieved, increase the output current by connecting the parallel resistance shown in Fig. 4. Use the electrical switch and configure the switching time accordingly. Show the transient response.

6. With the modified duty cycle of task 4, reduce the input voltage to 450V. Check the new output voltage. Again, modify the duty cycle to obtain the desired output voltage.

7. **Comment all your results**. Why the output voltage is affected when increasing the load current? What happens with the inductor current and semiconductor currents?